

1. Linux Basics

1.1 What is Linux?

Write a simple explanation in your own words:

Linux is an open-source operating system/kernel that is widely used in servers, cloud computing, development environments, and other systems.

1.2 Why do we need Linux?

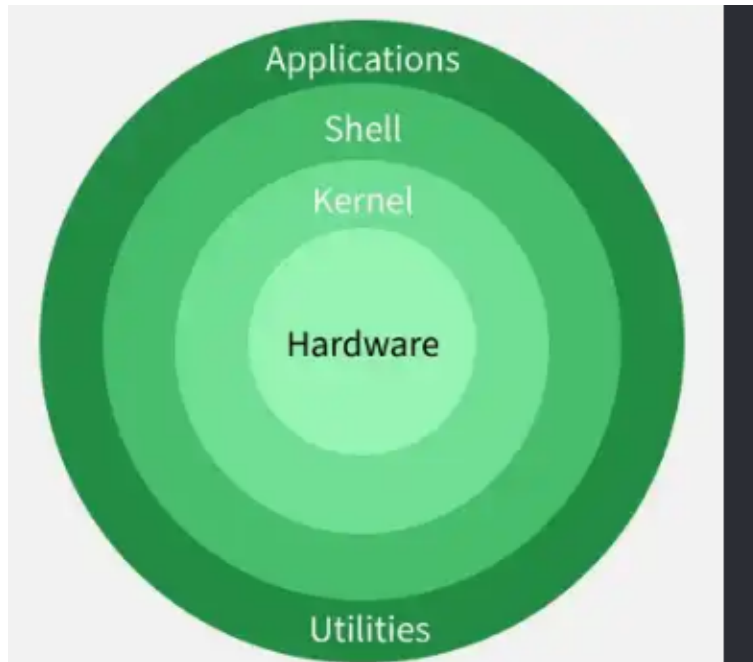
- Open source
- Secure and stable
- Lightweight
- Good for servers
- Widely used in cloud computing
- Powerful command-line interface
- Useful for DevOps and containerization

1.3 Linux Architecture

The Kernel: Technically, "Linux" refers only to the kernel. It is the core heart of the OS that directly controls the computer's CPU, memory, and peripheral devices.

Distributions (Distros): Because it is open-source, developers bundle the kernel with various tools and software to create complete, functional operating systems called "distributions". Popular distros include Ubuntu, Debian, and Linux Mint.

The Shell (CLI): Linux relies heavily on a command-line interface called a shell (like Bash), allowing users to execute powerful text commands to run the system efficiently. Modern versions also include full graphical user interfaces (GUIs). [1, 2, 3]



2. Linux Commands

Create a table for the commands you learned.

Command	Purpose	Example
<code>pwd</code>	Shows current directory	<code>pwd</code>
<code>ls</code>	Lists files/folders	<code>ls</code>
<code>cd</code>	Changes directory	<code>cd Documents</code>
<code>mkdir</code>	Creates a directory	<code>mkdir test</code>
<code>touch</code>	Creates a file	<code>touch app.py</code>
<code>cat</code>	Displays file content	<code>cat app.py</code>
<code>cp</code>	Copies files	<code>cp a.txt b.txt</code>
<code>mv</code>	Moves/renames files	<code>mv a.txt test/</code>
<code>rm</code>	Removes files	<code>rm a.txt</code>
<code>rmdir</code>	Removes empty directory	<code>rmdir test</code>
<code>clear</code>	Clears terminal	<code>clear</code>
<code>whoami</code>	Shows current user	<code>whoami</code>

Example

```
mkdir docker-projectcd docker-projecttouch app.pyls
```

What happened?

1. Created a directory called `docker-project`
2. Entered the directory
3. Created `app.py`
4. Listed the contents

This makes your documentation much more useful than just listing commands.

3. Docker Images

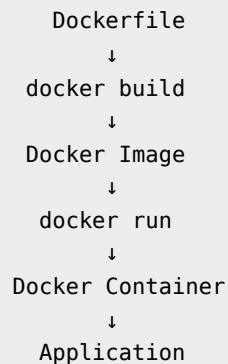
Since you specifically learned Docker Images, give this its own major section.

What is a Docker Image?

A Docker image is a read-only template/package containing the application, required dependencies, libraries, and instructions needed to create a container.

Image → Container

This is an important diagram:



Docker Image vs Container

Docker Image	Docker Container
Template/package	Running instance
Used to create containers	Runs the application
Read-only	Has a writable layer
Created using <code>docker build</code>	Created/started using <code>docker run</code>

Useful commands

```
docker images
```

Shows the Docker images available on your system.

```
docker pull python
```

Downloads an image from Docker Hub.

```
docker build-t myapp .
```

Creates an image using a Dockerfile.

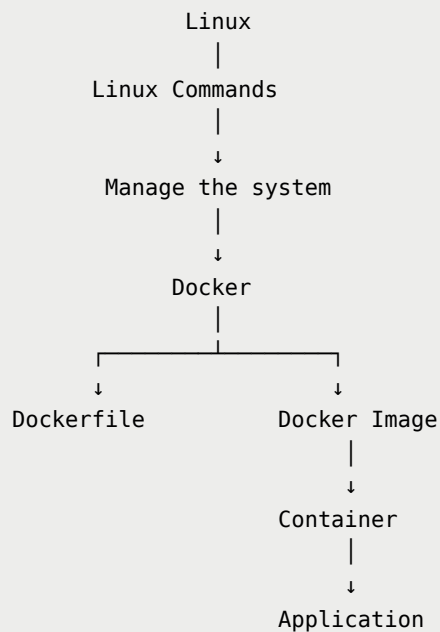
```
docker run myapp
```

Creates and runs a container from the image.

```
docker ps
```

Shows running containers.

5.How Everything Works ?



6.Summary

What I learned

- Basic Linux concepts
- Linux file and directory management
- Common Linux commands
- Basics of Docker
- Difference between an image and a container
- How Docker images are created and used
- Basic Docker commands
- Relationship between Dockerfile, image, and container.